

FPGA Implementation of High-Speed LDPC Decoder with Cross-Layer Scheduling for CV-QKD

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Abstract—This paper addresses the high-throughput requirements of information reconciliation in continuous-variable quantum key distribution (CV-QKD) systems by proposing an FPGA-based quasi-cyclic low-density parity-check (QC-LDPC) decoder implementation utilizing cross-layer scheduling. Through a pre-processing approach involving base matrix row reordering and data merging, the scheme effectively resolves memory access conflicts in layered decoding, enabling conflict-free parallel processing. Operating at 250 MHz, the decoder achieves a throughput of 100.5 Mbps, representing a 20% improvement over conventional approaches while maintaining equivalent frame error rate performance.

Index Terms—CV-QKD, LDPC, layered decoding, FPGA

I. INTRODUCTION

With the rapid advancement of quantum computing, many existing public-key cryptosystems face the threat of being cracked, undermining the foundational security of information transmission. Against this backdrop, quantum key distribution (QKD) [1]—a secure communication technology based on quantum mechanical principles—offers information-theoretically provable security, positioning it as a forward-looking solution to future security challenges. Among the two primary implementation forms of QKD, continuous-variable QKD(CV-QKD) [2] has garnered extensive research attention due to its significant advantages [3]. These include lower cost, higher integration, and greater potential for key rates in short-distance metropolitan area networks, achieved by leveraging standard coherent optical communication devices.

The actual performance of CV-QKD systems is largely constrained by the efficiency of their post-processing procedures. This process encompasses critical steps such as basis sifting, identity authentication, information reconciliation(IR) [4], and privacy amplification. Among these, the IR phase is responsible for correcting errors introduced by quantum

channel noise and potential eavesdropping, with its processing speed directly determining the system's ultimate key throughput. Since CV-QKD systems typically operate under extremely low signal-to-noise ratios(SNR), IR requires error-correcting codes approaching the Shannon limit. Low-Density parity-check (LDPC) codes [5] have become the preferred solution for IR due to their exceptional error correction capabilities. However, achieving high error correction performance necessitates long codewords and extensive iterative decoding computations, posing significant challenges to IR throughput and latency [6].

Traditional IR solutions rely on central processing units (CPU) or graphics processing units (GPU) for software acceleration [7]. While GPUs can significantly boost throughput through massively parallel computing, their instruction-based architecture still incurs inherent power consumption and latency overhead. In contrast, field-programmable gate arrays (FPGA) offer a more compelling hardware platform for implementing high-speed, low-latency LDPC decoders due to their high parallelism, customizable pipeline structures, and lower power consumption [8]. Particularly, implementing LDPC codes with quasi-cyclic properties on FPGAs can significantly simplify the design complexity of encoders and decoders by leveraging their regular structures [9], [10].

However, while quasi-cyclic LDPC (QC-LDPC) codes can achieve high data throughput in hardware implementations using a pipelined decoding architecture based on Layered Belief Propagation (LBP), significant complexity remains [11], [12]. This complexity arises because parity nodes within adjacent decoding layers share connections to certain variable nodes, making access to these shared nodes a bottleneck. On one hand, this characteristic limits the layer-by-layer parallel decoding capability of QC-LDPC codes. On the other

hand, when employing a pipelined decoding architecture, it significantly increases the risk of memory access conflicts. Specifically, memory conflicts arise if the information needed for processing the current row is not yet available from the preceding row's write operation. To prevent this read-write hazard, intentional idle cycles must be inserted to delay the read operation, which subsequently increases decoding latency and reduces the decoder's throughput.

This paper proposes a cross-layer scheduling strategy based on QC-LDPC encoding characteristics, along with a processing and storage scheme for the basis matrix, aiming to reduce the time required to obtain position indices and enhance parallelism. Compared to layer-by-layer updates, this approach delivers lower latency and higher throughput. Additionally, we describe a scheme for implementing a high-throughput decoder on an FPGA. Experimental results demonstrate that this approach can increase decoder throughput by 20% without compromising decoding performance.

The remainder of this paper is structured as follows. Section II begins with a brief background on QC-LDPC codes and the L-BP algorithm. Section III proceeds to detail the architecture of the proposed decoder. Section IV offers the simulation and implementation results, and the work concludes with Section V.

II. PRELIMINARIES

A. Quasi-Cyclic LDPC Codes

The decoding process in a standard LDPC architecture is facilitated by the iterative message exchange between variable and check node units over a Tanner graph based interconnect network, as depicted in Fig. 1. The unstructured interconnect introduced by purely-random parity-check matrices (PCM) between variable and check nodes hinders scalability in FPGA implementations, as it generates disordered memory access patterns and complicates routing. Quasi-cyclic codes offer a solution to these decoder challenges by adopting a matrix structure that is both highly regular and sufficiently random [13].

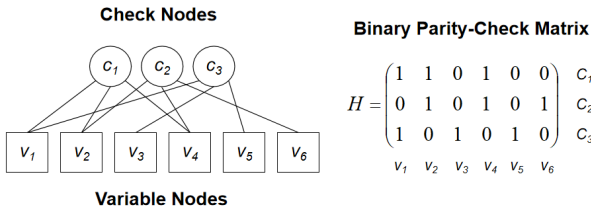


Fig. 1. LDPC Tanner graph and its corresponding PCM.

As illustrated in Fig. 2, the PCM of a QC code is composed of an array of $q \times q$ submatrices. These submatrices are either cyclically-shifted identity matrices or zero matrices of the same size. The expansion factor q in this structure governs the critical trade-off between the decoder's implementation

complexity and its error-correction capability [14]. The error-correction capability is improved by a small q , which increases the matrix's randomness. In contrast, decoder complexity is reduced by a large q , though this comes with a sacrifice in performance [15].

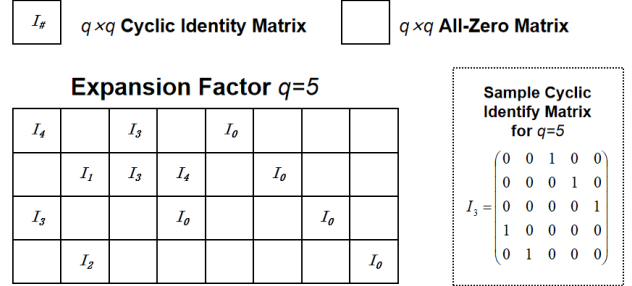


Fig. 2. Example of a PCM with $q=5$.

B. Layered Decoder Architecture

Currently, most decoders employ the Minimum Sum Algorithm (MSA) [16]. Although MSA is more hardware-friendly, it incurs significant decoding efficiency losses compared to the Belief Propagation (BP) algorithm. Furthermore, compared to flood scheduling, the layered BP (L-BP) scheduling implemented on FPGAs requires only one type of computational node unit (CNU). To address the computational complexity of algorithms in the probability domain, we introduce the log-likelihood ratio (LLR) into the decoding process. The detailed process is explained as follows.

Step (a): Initialization, check node updates, and variable node updates are all calculated based on

$$L(P_j) = \ln \frac{P_j(0)}{P_j(1)}. \quad (1)$$

Step (b): The calculation of the LLR message transmitted from the variable node to the check node is given by

$$L_{qji}^{(l,k)} = \begin{cases} L(P_j), & l = 1 \text{ and } k = 1 \\ L_{Q_j}^{(l-1,k)} - L_{r_{ij}}^{(l-1,k)}, & l \neq 1 \text{ and } k = 1, \\ L_{Q_j}^{(l,k-1)} - L_{r_{ij}}^{(l,k-1)}, & \text{else} \end{cases} \quad (2)$$

Where l denotes the number of layers in layered decoding, corresponding to the rows of the PCM, k denotes the iteration number, i is the index of the check node, and j is the index of the variable node.

Step (c): The update of the variable node LLR message transmitted to the variable node is given by

$$L_{r_{ij}}^{(l,k)} = \left(\prod_{j' \in V(i) \setminus j} \text{sgn}(L_{qj'i}^{(l,k)}) \right) \cdot \Psi \left(\sum_{j' \in V(i) \setminus j} \Psi(L_{qj'i}^{(l,k)}) \right), \quad (3)$$

$$\Psi(x) = -\log(\tanh \frac{|x|}{2}). \quad (4)$$

Step (d):The posterior LLR is updated by

$$L_{Q_j}^{(l,k)} = L_{q_{ji}}^{(l,k)} + \sum_{i \in C(j)} L_{r_{ij}}^{(l,k)}. \quad (5)$$

Step (e):Hard decision decoding is performed by

$$c = \begin{cases} 1, & L_{Q_j}^{(l,k)} < 0 \\ 0, & \text{else} \end{cases}. \quad (6)$$

Step (f):If the maximum number of iterations is reached or $H^T \cdot c = S$ is satisfied, the decoding process terminates.where S denotes the syndrome sent by Bob to aid in decoding during IR.

III. FPGA ARCHITECTURE

This section provides a comprehensive overview of the FPGA-based QC-LDPC decoder architecture, designed to enhance decoding speed and improve overall system throughput.

A. The Memory Strucure and Datapath

The FPGA-based partially-parallel decoder shown in Fig. 3 employs three types of memory: PCM_RAM, LLR_RAM, and Lc2v_RAM. Data read and write operations are efficiently managed solely through address switching.

In the design of a layered decoder, the intra-layer computation comprises three steps: the calculation of variable node LLR messages, the calculation of check node LLR messages, and the calculation of posterior LLR messages. The calculation of LLR messages at check nodes is a subtraction operation, which is simpler than the flooding strategy. In contrast, the computation of the LLR message for variable nodes involves multiplying the sign bit by the value of the $\Psi(x)$ function. To simplify the decoder design, the complex function $\Psi(x)$ is quantized. The computation of its function values is thereby streamlined into a convenient lookup step, which significantly increases the processing speed while reducing hardware resource consumption. We implement the sign multiplication operation by employing an XOR module, and we realize the accumulation operation by utilizing a Sigma module. The posterior LLR message is the sum of the current layer's variable node LLR message and the check node LLR message.

B. Preprocessing of The PCM Matrix

To simplify the decoder design and reduce hardware complexity, this paper preprocesses the base matrix H before decoding to obtain a regularized matrix H' . The procedure is as follows: First, the non-negative values of H are extracted to form a numerical matrix, termed the Extraction Matrix. This matrix contains the shift values representing the right cyclic shift offsets for the circulant permutation matrices. The cyclic shift module utilizes these values to perform cyclic shifts on the LLRs.

Subsequently, address values are derived from the Extraction Matrix. An Offset Matrix (Offset1) is created based on the values in the Extraction Matrix. Concurrently, an Address Matrix (Addr1) is generated according to its corresponding

row and column indices, which is crucial for determining the addresses of the LLRs in the memory modules.

Finally, the offset and address matrices are regularized. Specifically, the Offset Matrix is padded with the value z , while the Address Matrix is padded with 0.

Fig. 4(a) provides a visual description of the base matrix preprocessing procedure.

C. Cross-layer Scheduling Approach

Traditional layered decoding architectures suffer from a significant performance bottleneck due to sequential timing overhead during node updates. The core issue lies in the stringent data access dependency that enforces a layer-by-layer serial execution flow: the processor must read the corresponding Log-Likelihood Ratios (LLRs) for computation layer by layer, based on the address and offset information obtained from base matrix preprocessing. Only after all node updates in the current layer are completed and the results written back to memory can the processor proceed to read the latest LLR values for the next layer according to its address information. This constraint arises primarily because adjacent layers often contend for the same memory addresses, forcing the update of the current layer to wait for the full completion of the preceding layer's computations, thereby introducing numerous idle wait cycles into the process. This strict serial dependency results in substantial timing overhead.

To overcome this fundamental challenge, this paper proposes a parallel scheduling scheme based on row reordering and data merging of the PCM. As illustrated in Fig.4(b), the proposed method first reorders the rows of the PCM, grouping rows with non-conflicting addresses together. Subsequently, the data corresponding to these conflict-free rows are merged for storage, and a synchronous merging operation is applied to the associated offset matrix. This design enables the system to parallelly read and update the LLR values associated with multiple non-conflicting row addresses during node processing. Consequently, it entirely eliminates the temporal overhead inherent in the traditional inter-row serial data access routine. Experimental results confirm that the proposed scheme achieves a significant reduction in the total clock cycles required for the decoding process.

IV. EXPERIMENTAL RESULTS

The QC-LDPC decoder we proposed has been implemented and tested on the Xilinx Vertex-7 FPGA platform. Experiments employed a QC-LDPC code compliant with the ATSC 3.0 standard, featuring a length of 64,800 bits and a code rate of 3/15.

In FPGA-based LDPC decoder design, achieving high throughput stands as a fundamental advantage over software implementations. To precisely quantify and evaluate the actual performance of the decoder, the throughput Γ is widely adopted as a key performance metric. This metric Γ can be defined and calculated using the following formula:

$$\Gamma = \frac{f_{clk} \times N}{T_{in} + T_{iter} \times I_{max}}, \quad (7)$$

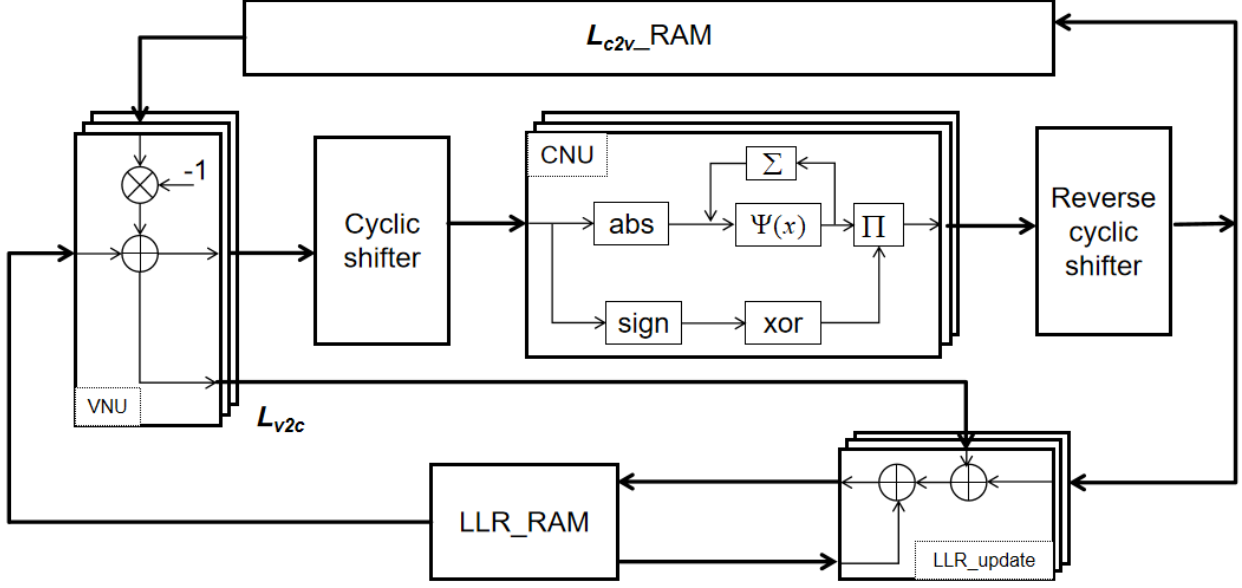


Fig. 3. Architecture of an FPGA-Based LDPC decoder.

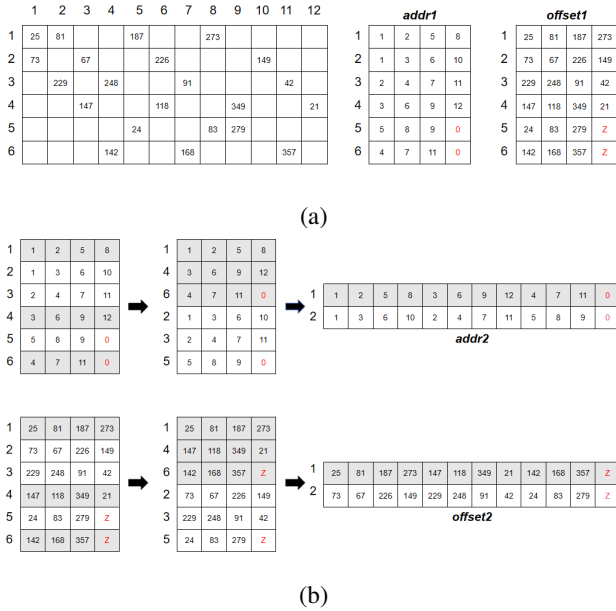


Fig. 4. (a) Base graph matrix example with its original processing schedule. (b) An instance of scheduling and reorganization during the decoding algorithm optimization.

where f_{clk} is the system clock frequency, N is the code length, and I_{max} is the maximum number of iterations. Parameter T_{iter} denotes the number of clock cycles required per iteration, and T_{in} represents the input data transmission delay.

With the clock frequency configured at 250 MHz, the

proposed decoder achieves a maximum throughput of 100.5 Mbps. Compared to the baseline architecture without cross-row scheduling of the base matrix, this represents a throughput improvement of approximately 20%, demonstrating the effectiveness of the proposed scheduling strategy in enhancing data throughput capability. Additionally, the overall resource utilization of the decoder is shown in Table I. The core advantage of our scheme lies in achieving a lower code rate than that in [17], which is crucial for improving decoding performance. However, this lower code rate incurs inevitable trade-offs: the number of nodes to be processed increases significantly. Meanwhile, to ensure fixed-point precision, we adopt a larger bit width, leading to increased consumption of LUTs and FFs. In addition, the relatively high number of basic iterations results in a slightly lower throughput. Nevertheless, these overheads remain within an acceptable range, and the overall results still meet the latest requirements in CV-QKD.

To comprehensively evaluate the overall performance of the proposed scheduling scheme, this study conducts a comparative analysis of the frame error rate (FER) of the decoder before and after optimization. Experimental results demonstrate that while significantly improving the throughput, the proposed cross-layer scheduling scheme has a negligible impact on the system's FER performance, with its performance curve largely overlapping with that of the pre-optimized implementation, as illustrated in the accompanying fig.5.

V. CONCLUSION

This study successfully designed and implemented a high-throughput QC-LDPC decoder suitable for CV-QKD systems.

TABLE I
RESOURCE UTILIZATION OF THE LAYERED DECODER

	[17]	This work
FPGA device	Kintex-7(XC7k325tfg900-2)	Vertex-7(XC7VX690T)
Algorithm	MSA	BP
Schedule	Layered	Layered
Code rate	1/4	3/15
Code length	64800	64800
Width of fixed-point numbers(bits)	9	19
Throughput(Mbps)	150	100.5
LUT	115864	189204(43.6%)
FF	57931	599381(69.1%)
BRAM	180	360.5(24.5%)

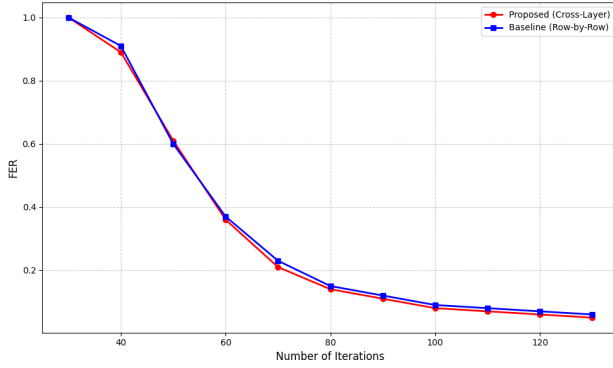


Fig. 5. FER versus the number of iterations for the two scheduling schemes under a fixed Signal-to-Noise Ratio (SNR) condition.

By analyzing the timing bottlenecks in traditional layered decoding architectures, an innovative cross-layer scheduling scheme based on base matrix row rearrangement and data merging was proposed. This scheme effectively addresses the inter-layer address conflict issue in LDPC decoding during the information reconciliation phase of CV-QKD systems, enabling partial parallel processing. Experimental results demonstrate that at a clock frequency of 250 MHz, the decoder achieves a maximum throughput of 100.5 Mbps, representing approximately 20% improvement compared to conventional schemes while maintaining equivalent error correction performance. This efficient decoder architecture is particularly well-suited for the stringent real-time post-processing requirements of CV-QKD systems and can effectively enhance the system's final secret key generation rate.

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REFERENCES

- [1] N. Gisin, G. Ribordy, W. Tittel, and H. Zbinden, "Quantum cryptography," *Rev. Mod. Phys.*, vol. 74, no. 1, pp. 145–195, Jan. 2002.

- [2] E. Diamanti and A. Leverrier, "Distributing secret keys with quantum continuous variables: principle, security and implementations," *Entropy*, vol. 17, no. 9, pp. 6072–6092, Sep. 2015.
- [3] B. Ye, J. Liu, and H. Li, "FPGA accelerated adaptive LDPC-based quantum error correction by bitwise pipeline parallelism," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, 2025, pp. 1–5.
- [4] X.-Q. Jiang, P. Huang, D. Huang, D.-K. Lin, and G.-H. Zeng, "Secret information reconciliation based on punctured low-density parity-check codes for continuous-variable quantum key distribution," *Phys. Rev. A*, vol. 95, no. 2, p. 022318, Feb. 2017.
- [5] R. Gallager, "Low-density parity-check codes," *IRE Trans. Inf. Theory*, vol. 8, no. 1, pp. 21–28, Jan. 1962.
- [6] S.-S. Yang, Z.-G. Lu, and Y.-M. Li, "High-speed post-processing in continuous-variable quantum key distribution based on FPGA implementation," *J. Lightw. Technol.*, vol. 38, no. 15, pp. 3935–3941, Aug. 2020.
- [7] D. Oh and K. K. Parhi, "Low-complexity switch network for reconfigurable LDPC decoders," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 18, no. 1, pp. 85–94, Jan. 2010.
- [8] M. Paramaiyappan and S. P. Rajan, "Efficient FPGA implementation of a high-performance, low-complexity LDPC decoder for next-gen communication systems," in *Proc. Int. Conf. Frontier Technol. Solutions (ICFTS)*, 2025, pp. 1–9.
- [9] A. Tanaka, M. Fujiwara, K. Yoshino, S. Takahashi, Y. Nambu, A. Tomita, S. Miki, T. Yamashita, Z. Wang, M. Sasaki, and A. Tajima, "High-speed quantum key distribution system for 1-Mbps real-time key generation," *IEEE J. Quantum Electron.*, vol. 48, no. 4, pp. 542–550, Apr. 2012.
- [10] Z. Wang, Z. Xie, K. Gong, et al., "High-performance QC-LDPC layered decoder based on shortcut updating," *IEEE Signal Processing Letters*, vol. 31, pp. 596–600, 2024.
- [11] V. L. Petrović, M. M. Markovic, D. M. El Mezeni, and V. R. Stojanović, "Flexible high throughput QC-LDPC decoder with perfect pipeline conflicts resolution and efficient hardware utilization," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 67, no. 12, pp. 5454–5467, Dec. 2020.
- [12] D. E. Hocevar, "A reduced complexity decoder architecture via layered decoding of LDPC codes," in *Proc. IEEE Workshop Signal Process. Syst.*, 2004, pp. 107–112.
- [13] M. P. C. Fossorier, "Quasicyclic low-density parity-check codes from circulant permutation matrices," *IEEE Trans. Inf. Theory*, vol. 50, no. 8, pp. 1788–1793, Aug. 2004.
- [14] M. Milicevic, C. Feng, L. M. Zhang, and P. G. Gulak, "Key reconciliation with low-density parity-check codes for long-distance quantum cryptography," *arXiv:1702.07740*, Feb. 2017.
- [15] D. Chang, G. Wang, G. Yan, Y. Chen, and X. Lin, "An optimization model for offline scheduling policy of low-density parity-check codes," *IEEE Commun. Lett.*, vol. 28, no. 8, pp. 1740–1744, Aug. 2024.
- [16] M. P. C. Fossorier, M. Mihaljevic, and H. Imai, "Reduced complexity iterative decoding of low-density parity check codes based on belief propagation," *IEEE Trans. Commun.*, vol. 47, no. 5, pp. 673–680, May 1999.
- [17] E. A. Likhobabin, A. A. Ovinnikov, R. S. Goriushkin, P. B. Nikishkin, and E. I. Khokhryakov, "High throughput FPGA implementation of LDPC decoder architecture for DVB-S2X standard," in *Int. Conference Inf. Control Commun. Technol. (ICCT)*, 2022, pp. 1–4.